Candidate Seat No:………………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, AMT)

March-April 2018

ME784/ME734 Computer Aided Engineering

Date: 05.04.2018, Thursday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator is allowed.

SECTION- I

Q1	**(a)**	Using a simplified description of process of design explain where computational aids might be assistance to the designer.	**5**
	(b)	Prepare the detailed specifications for a typical CAD workstation with latest hardware & state the different commercially available CAE software.	**6**
Q2	**(a)**	Briefly explain flat panel displays.	**3**
	(b)	Explain the working principle of plotter.	**3**
	(c)	Using DDA line algorithm, find the pixel positions between end points (6, 9) and (11, 12).	**6**
		OR	
Q2	**(a)**	Why can't a vector scan display produce as many colours as a raster scan display?	**3**
	(b)	Explain interpolated curves and approximated curves.	**3**
	(c)	Using Bresenham's line algorithm, find the pixel positions between end points (0, 2) and (4, 5).	**6**
Q3	**(a)**	What is transformation? Compare geometric transformation with coordinate transformation.	**3**
	(b)	Prove that 3D rotations are not commutative.	**3**
	(c)	A square having end points A(1,1), B(6,1), C(6,6), D(1,6) is rotated by 50° clockwise direction keeping point (6,1) fixed. Find the final coordinates.	**6**
		OR	
Q3	**(a)**	Explain 2-D transformation for shearing in x – axis & y - axis.	**3**
	(b)	What are the advantages of homogenous coordinate system?	**3**
	(c)	Find the reflection of point (5, 5) about a line $y = 3x + 8$.	**6**

SECTION- II

Q4 (a) Compare cubic Bezier curve with Hermite cubic curve. **4**

(b) Explain the principle of Rapid Prototyping, its advantages and applications. **3**

(c) Explain in brief reverse engineering? What are active scanner and passive scanner? **3**

Q5 Derive weak form and finite element formulation for following governing differential equation using linear element. **7**

$$C_1 \frac{d^2u}{dx^2} - C_2 u = 0$$

OR

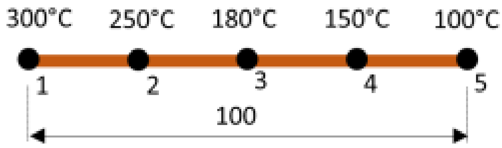
Q5 Discuss application of finite element method in solving an engineering problem. **7**

Q6 **Attempt any *three*:** **18**

(a) Discuss the advantages, limitations and applications of Finite Difference and Finite Element Methods.

(b) Derive the finite difference formula to compute $\frac{dT}{dx}$ using central difference scheme.

(c) In a rod having 100 mm length, subjected to heat transfer, equilibrium temperatures at 5 equidistance nodes are shown in figure below. Determine the temperature at point 30 mm and 90 mm from the left wall considering the linear elements.



(d) One dimensional deformation inside the one meter long cast steel rod is governed by following equation. Find a three parameter solution using Galerkin Method.

$$\frac{d^2T}{dx^2} = 2x$$

Subjected to following boundary conditions.

At $x = 0$, $T = 10°C$; and at $x = 1$, $T = 100°C$

(e) Briefly explain solution algorithm for Jacobi method using an illustration.

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